

Quiz Week 4



Cycles

1. Factor each of the following permutations into disjoint cycles.

$$(a) \sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 3 & 8 & 9 & 5 & 2 & 1 & 6 & 4 & 7 \end{pmatrix} \quad (b) \tau = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 6 & 4 & 8 & 9 & 3 & 1 & 7 & 5 & 2 \end{pmatrix}$$

$$(c) \gamma = (1 \ 2 \ 3 \ 4 \ 5)(6 \ 7)(1 \ 3 \ 5 \ 7)(1 \ 6 \ 3)$$

Solution:

$$(a) \sigma = (1 \ 3 \ 9 \ 7 \ 6)(2 \ 8 \ 4 \ 5).$$

$$(b) \tau = (1 \ 6)(2 \ 4 \ 9)(3 \ 8 \ 5).$$

(c) Observe that γ is a product of cycles but they are not disjoint. Thus, in order to find the factorization of γ into disjoint cycles, we need to write it first in two-row matrix notation.

$$\gamma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 7 & 3 & 4 & 5 & 6 & 1 & 2 \end{pmatrix} = (1 \ 7 \ 2 \ 3 \ 4 \ 5 \ 6)$$

2. Let $\sigma = (1 \ 8 \ 7 \ 4)(3 \ 6 \ 7 \ 5 \ 9)$, $\tau = (1 \ 3 \ 5 \ 7)(4 \ 1 \ 9)$,
 $\alpha = (1 \ 4 \ 5)(3 \ 7)(6 \ 8 \ 2)$, $\beta = (6 \ 1 \ 4 \ 8)(2 \ 3 \ 4 \ 5)(1 \ 2 \ 4 \ 9 \ 3)$
be permutations in S_9 .

Mark the correct answers; when incorrect provide a correct answer.

(a) $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 8 & 2 & 6 & 1 & 9 & 4 & 5 & 7 & 3 \end{pmatrix}$

(b) $\tau = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 9 & 2 & 5 & 3 & 7 & 6 & 1 & 8 & 4 \end{pmatrix}$

(c) $\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 4 & 6 & 7 & 5 & 1 & 8 & 3 & 2 & 9 \end{pmatrix}$

(d) $\beta = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 4 & 5 & 1 & 8 & 2 & 1 & 7 & 6 & 3 \end{pmatrix}$

Solution: (a), (b) and (c) are correct, while (d) is incorrect; we will see that $\beta(1) = 3$. Let $\beta_1 = (1 \ 2 \ 4 \ 9 \ 3)$, $\beta_2 = (2 \ 3 \ 4 \ 5)$ and $\beta_3 = (6 \ 1 \ 4 \ 8)$. Notice that $\beta = \beta_3\beta_2\beta_1$. We have the following:

$$1 \xrightarrow{\beta_1} 2 \xrightarrow{\beta_2} 3 \xrightarrow{\beta_3} 3$$

$$2 \xrightarrow{\beta_1} 4 \xrightarrow{\beta_2} 5 \xrightarrow{\beta_3} 5$$

$$3 \xrightarrow{\beta_1} 1 \xrightarrow{\beta_2} 1 \xrightarrow{\beta_3} 4$$

$$4 \xrightarrow{\beta_1} 9 \xrightarrow{\beta_2} 9 \xrightarrow{\beta_3} 9$$

$$5 \xrightarrow{\beta_1} 5 \xrightarrow{\beta_2} 2 \xrightarrow{\beta_3} 2$$

$$6 \xrightarrow{\beta_1} 6 \xrightarrow{\beta_2} 6 \xrightarrow{\beta_3} 1$$

$$7 \xrightarrow{\beta_1} 7 \xrightarrow{\beta_2} 7 \xrightarrow{\beta_3} 7$$

$$8 \xrightarrow{\beta_1} 8 \xrightarrow{\beta_2} 8 \xrightarrow{\beta_3} 6$$

$$9 \xrightarrow{\beta_1} 3 \xrightarrow{\beta_2} 4 \xrightarrow{\beta_3} 8$$

Putting altogether we have that

$$\beta = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 3 & 5 & 4 & 9 & 2 & 1 & 7 & 6 & 8 \end{pmatrix}$$